Worksheet 8.2

Changing states

Student: ..... Class: .....

Melting and boiling

The melting point of a substance is the temperature at which it will change from being a solid to being liquid, while the boiling point is the temperature at which a substance changes from a liquid state into a gas state. Some substances do not actually have a melting point because they go directly from a solid form to a gas. This is called sublimation.

The table below shows the various melting and boiling points and sublimation temperatures for different substances.

Substance	Melting point (°C)	Boiling point (°C)	Sublimation temperature (°C)
Oxygen	–220	–183	–
Nitrogen	–210	–196	–
Carbon dioxide	–	–	–78.5
Water	0	100	–
Table salt	804	1413	–
Lead	621	1749	–
Gold	1065	2856	–
Naphthalene (mothballs)	–	–	48

1. What state (solid, liquid or gas) would each of the following be in?

(a) Nitrogen at –200 °C

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(b) Carbon dioxide at 20 °C

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(c) Table salt at 1400 °C

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2. Which of the substances in the table would be solids at 1200 °C?

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3. Which of the substances in the table would be liquids at 900 °C?

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